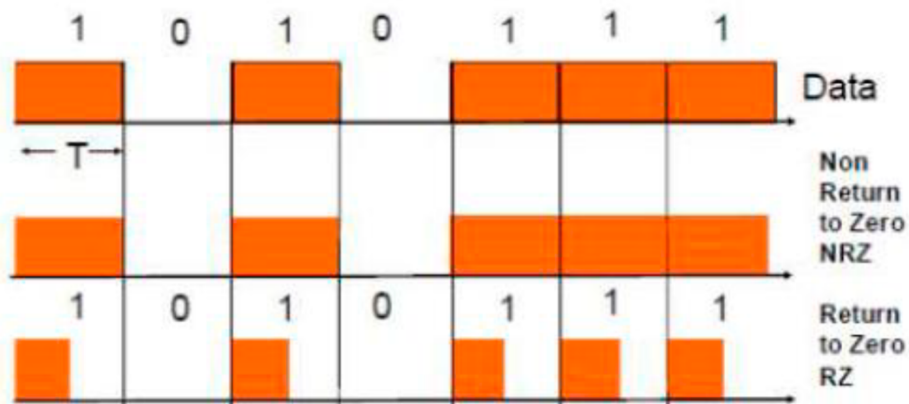


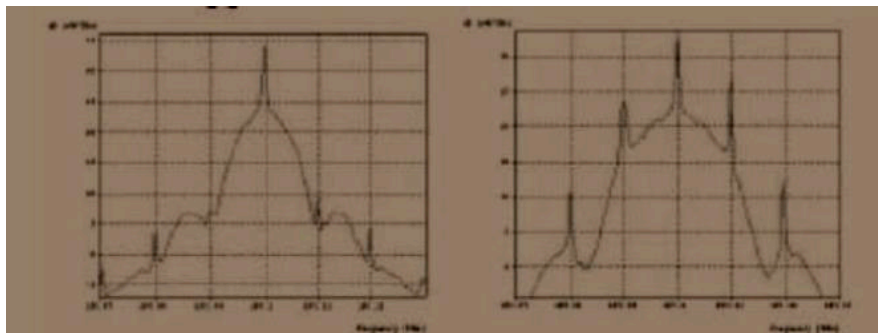
EE558 HW4

Problem 1:

Return-to-zero (RZ) describes a line code used in telecommunications signals in which the signal drops (returns) to zero between each pulse. In contrast, non-return-to-zero (NRZ) formats permit constant optical intensity over several consecutive bits.



The following is 10 G/s optical signal spectra of RZ and NRZ modulation formats.



- Which one is for RZ, and which one is for NRZ? Why?
- Which one is more tolerant to dispersion in a fiber link?
- How far away from each other the peaks in the picture on the right side are supposed to be?

Problem 2:

- Draw the general scheme of heterodyne receiver and derive the output current.
- Is it necessary to have $P_{LO} \gg P_s$? If so, explain why?
- Suppose P_{LO} is increased by 3 dB. How does the output current change?
- If we increase P_{LO} by 3 dB and decrease P_s by 2 dB. how does the output current change?

Problem 3:

If nonlinear phase shift induced by self-phase modulation is given by $\phi_{NL} = \gamma \cdot P_{in} \cdot L$, where nonlinear coefficient is defined as $\gamma = \frac{2\pi n_2}{A_{eff}\lambda}$, where n_2 is the nonlinear index coefficient, λ is the free space wavelength, and A_{eff} is the effective core area. A 1550 nm continuous wave signal with 6 dBm power is launched into a fiber with $50 \mu\text{m}^2$ effective mode area. After what fiber length L would be induced nonlinear phase shift becomes 2π ? Assume $n_2 = 2.6 \times 10^{-20} \text{ m}^2/\text{W}$. Neglect fiber loss.

Problem 4:

- (a) Compare 16-QAM and QPSK in terms of BER and spectral efficiency.
- (b) Consider a communication system in which the transmission link has very low noise and there is a limitation for provided bandwidth. Which modulation format do you suggest for such a system (16-QAM or QPSK)? Why?
- (c) If SNR level is low and the bandwidth of this system is not the issue, what is your preference in modulation format? Why?